



ESP32 RC CAR

Felipe Oscanoa

ECE 528

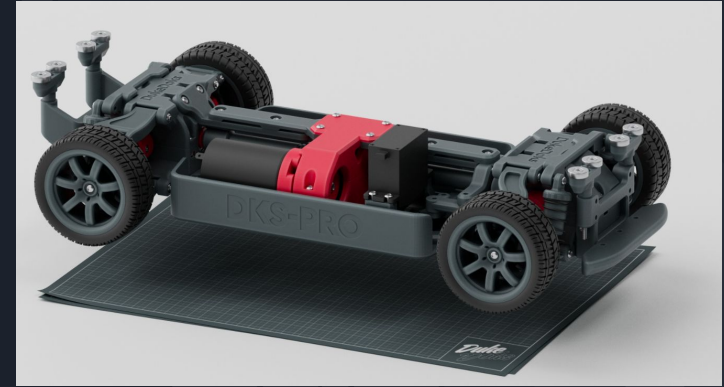
Project Overview:

- This system demonstrates wireless embedded control and real-time video feedback in a robotics application.
- The objective of this project is to design and implement a remotely controlled (RC) car using an ESP32 microcontroller and implementing an live recording of the drive using an ESP32 CAM.
- Last objective is to make an budget RC Car that is less than 100\$



System Architecture:

- 3D printed RC car structures was made as the main chassis for this project
- Main brain of the RC car is the ESP32 that is in charge of controlling the drivability of it by using an ps5 controller.
- Additional module is the ESP32 CAM that is used to get the live video while driving the RC Car
- Using different components like the 35T brushed motor, LM2596s Buck converter, Treehobby RC 60A Brushed ESC



Interfaces and Peripherals:

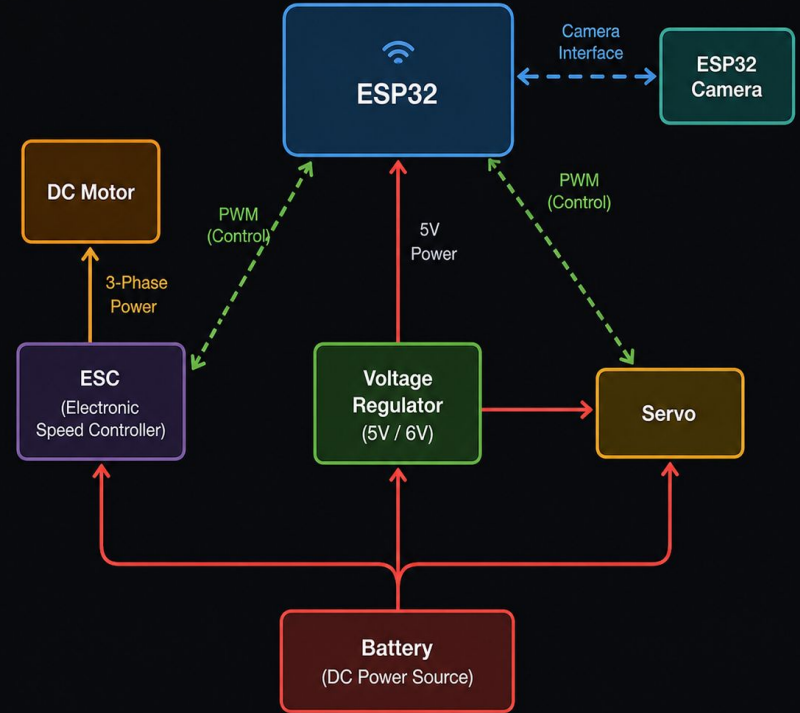
Bluetooth -> PS5 Controller Input (using MAC Address)

PWM -> Motor and Servo control (R2 and L2 and stick shift)

Timers -> PWM Signal Generation

GPIO -> Signal Connections (ESP32)

Wifi -> Camera video streaming (ESP32 CAM)



Coding Structure

- The code was made in Arduino IDE and the libraries of ESP32- Servo, ESP32-PS5, and Esp32 Espressif were used to start creating the code for this car.
- The code will consist of various loops and threshold for the inputs of the controller for the steering and moving forward/reverse.
- PWM signals on the servo were tuned so it will go on the desired range and not hit the walls of the chassis.
- PWM signals for the ESC were lowered and adjusted so it will not spin too fast and burn the plastic or crash and break, still because of the ESC limitation the reverse will always be slower than forward
- ESP32 CAM was setup using the library of Camera Web Browser and then modified so it will be generating its own Wifi and using the website that is generated using the IP address given we can see the live video of the camera. Although it is still laggy which could be caused from an unstable connection of the wifi on the ESP32 CAM or limitations on the connection with the camera

```
#include <ps5Controller.h>
#include <ESP32Servo.h>
```

```
if (r2 > 10 && l2 <= 10) {
  pulse = map(r2, 0, 255, ESC_FORWARD_MIN, ESC_FORWARD_MAX);
}
else if (l2 > 10 && r2 <= 10) {
  pulse = map(l2, 0, 255, ESC_REVERSE_MIN, ESC_REVERSE_MAX);
}
```

```
// ----- STEERING -----
int stickX = ps5.LStickX();

int servoPulse = mapSteering(stickX);
steering.writeMicroseconds(servoPulse);
```

```
// ----- Pins -----
const int ESC_PIN = 18;
const int SERVO_PIN = 19;

// ----- ESC -----
const int ESC_NEUTRAL = 1500;
const int ESC_FORWARD_MIN = 1550;
const int ESC_FORWARD_MAX = 1700;
const int ESC_REVERSE_MIN = 1450;
const int ESC_REVERSE_MAX = 1300;

// ----- SERVO -----
const int SERVO_MIN = 750;
const int SERVO_MAX = 1900;
const int SERVO_CENTER = 1350;

bool escArmed = false;
unsigned long armStart = 0;
```



PIN's Used

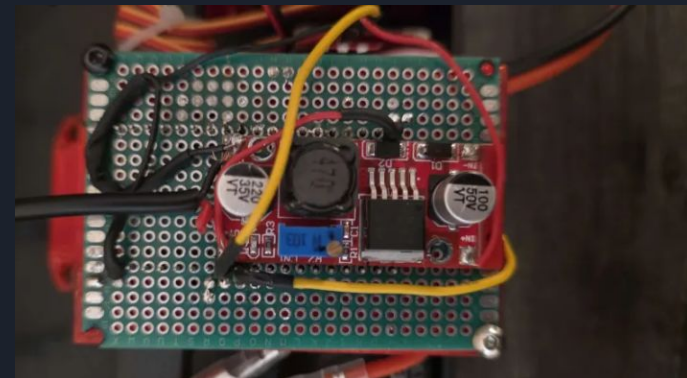
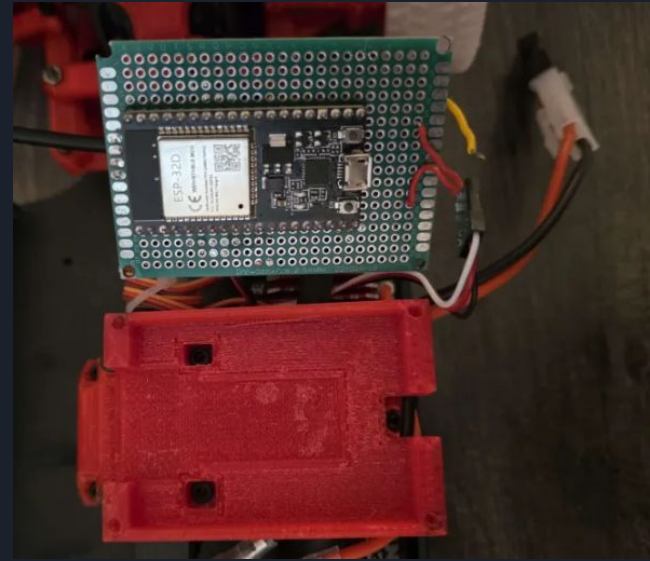
- Electronic Speed Controller (ESC) controlled by GPIO18 from the ESP32
- Servo motor was controlled by the GPIO19
- ESP32 CAM there is not present PIN used but the implementation of the WIFI generated by the module

ESP32 Pinout Diagram



≡ Verification and Testing:

- Testing consisted of getting the parts working using the ESP32 to see how they go with it.
- For the circuit soldered it was mostly just making sure the right voltage of 5V was being given out by the buck converter.
- Early testing for the code with the controller was to get the terminal to read the values of the controller when it was pressed or moved the stick shift and then using the values for threshold
- Same case for the servo the limitation go from 9 to 135 degrees for it to steer correctly
- Various parts were reprinted and modified since it was breaking or there was not enough clearance



Project Demonstration



https://youtu.be/UeRciERhonk?si=g5Z_SlrQulE9SANO

Conclusion and Improvements

- Successfully designed and implemented a wireless RC car using ESP32 and achieved live video camera
- Being able to use an ps5 controller to control this car
- Budget not going for more than 100 \$ if labor is not counted
- Improvement on the structure of the car since it is not driving staring and becomes lose on the wheels with time.
- Improve the ESP32 CAM to get a better video or change to a better set up.
- Upgrade the motor, and ESC to brushless to get more speed and reliability
- Upgrade from plastic to CNC machined parts especially for the wheels axles and differentials

